Abstract:

AI Personal Health Assistant and Medicine Management System

Effective healthcare management has become increasingly difficult due to busy lifestyles, irregular medication schedules, fragmented medical records, and limited awareness of preventive healthcare practices. Many individuals forget to take medicines on time, miss medical appointments, and rely on paper-based prescriptions and reports, resulting in poor medication adherence, delayed treatment, and inefficient health monitoring. These challenges highlight the need for an integrated digital solution that simplifies personal healthcare management while improving accessibility and user convenience.

The proposed **AI Personal Health Assistant and Medicine Management System** is a secure, intelligent, and user-friendly web application designed to assist users in managing their daily healthcare activities. The system enables users to maintain digital medical records, schedule medicines with automated reminder notifications, organize doctor appointments, and monitor essential health information through a centralized platform. To enhance user support, the application integrates Artificial Intelligence services through external APIs to provide symptom-based health guidance, general wellness recommendations, and responses to common health-related queries. The AI functionality is intended to provide informational assistance only and does not replace professional medical advice, diagnosis, or treatment.

The application will be developed using **Python** for backend development, **HTML, CSS, and JavaScript** for the frontend, **MySQL** for secure database management, and API integration for AI-enabled healthcare services. The project follows the **Software Development Life Cycle (SDLC)** to ensure a systematic, reliable, and scalable development process. By integrating healthcare management with AI-assisted guidance, the proposed system improves medication adherence, streamlines medical record management, promotes preventive healthcare awareness, and provides an efficient digital solution for supporting healthier daily routines.

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